IPL Player Auction Price Prediction System

Dataset - [IPL_auction]

🎯 Problem Statement

In the **Indian Premier League (IPL)**, franchises invest heavily in players during auctions. However, deciding how much a player is worth is complex and depends on multiple factors like **past performance, experience, and skills**.

The challenge is:

> **Can we predict a player's auction price based on their historical performance and attributes?**

🔍 Objective:

Build a **regression model** that predicts the **auction price of a player** using performance metrics and player characteristics.

💡 Business Impact:

- Helps teams make **data-driven bidding decisions**
- Avoids overpaying or undervaluing players
- Identifies undervalued high-potential players
- Improves overall team-building strategy

🎯 Target Variable

- **Column Name:** `Auction_Price` *(or similar price-related column)*
- **Description:** The price at which a player is sold in the auction
- **Type:** Numerical

👉 This is a **regression problem**

📊 Dataset Overview

- Features: Multiple player performance and profile features
- Type: Mixed (Numerical + Categorical)
- Domain: Sports Analytics / Cricket

📌 Column Information (Detailed Explanation)

(Note: Based on standard IPL auction datasets, adjust names slightly if needed in your file)

1. **Player_Name**
   - Type: Categorical (Identifier)
   - Description: Name of the player
2. **Age**
   - Type: Numerical
   - Description: Age of the player
3. **Role**
   - Type: Categorical
   - Values: Batsman, Bowler, All-rounder, Wicketkeeper
4. **Matches_Played**
   - Type: Numerical
   - Description: Total matches played
5. **Runs_Scored**
   - Type: Numerical
   - Description: Total runs scored
6. **Batting_Average**
   - Type: Numerical
   - Description: Average runs per innings
7. **Strike_Rate**
   - Type: Numerical
   - Description: Scoring rate (runs per 100 balls)
8. **Wickets_Taken**
   - Type: Numerical
   - Description: Total wickets taken
9. **Bowling_Average**
   - Type: Numerical
   - Description: Runs conceded per wicket
10. **Economy_Rate**

- Type: Numerical
- Description: Runs conceded per over

11. **International_Experience**

- Type: Binary (0/1)

- Description: Whether player has international experience

12. **Previous_Auction_Price**

- Type: Numerical
- Description: Price in previous auction

Target Column

13. **Auction_Price**

- Type: Numerical
- Description: Final price in auction
- Role: Target variable